

Spectral-ResiBit YOLO: A Frequency-Aware, Latent-Shared 1.58-bit Detector for Edge Intelligence

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Abstract—As Larger Foundation Models transition toward the 1.58-bit (ternary) era, compact computer vision models remain constrained by the “Capacity-Stability Dilemma.” While massive Large Language Models (LLMs) utilize parameter redundancy to absorb quantization noise, thin convolutional detectors like YOLO suffer from catastrophic gradient manifold shattering and codebook collapse when subjected to extreme quantization. We propose Spectral-ResiBit YOLO, a novel architecture that resolves the representational deficit of sub-2-bit quantization through three synergistic breakthroughs: (1) The Universal Spectral-ResiBit (USR) Cell, which resolves the mathematical conflict between Static Channel Splitting (SCS) and Base-5 Fusion via a Latent-Shared Manifold (LSM); (2) FFT-based Spectral Orthogonality, which forces expert specialization in the frequency domain; and (3) INT8 Residual Highways (RPA), which anchor the gradient manifold to prevent the “Muon Trap” divergence. Our forensic audit reveals that the interaction between the Muon optimizer and Straight-Through Estimators (STE) is the primary cause of mAP collapse in ternary networks. Experimental results on the COCO dataset demonstrate that Spectral-ResiBit YOLO recovers 94.0% of FP32 accuracy while achieving $13.78\times$ weight compression. Finally, we introduce a Hardware Manifesto for ARM NEON, detailing the Pack-Store-Load-Unpack-Compute (PSLUC) paradigm for bit-serial execution on edge targets like the Raspberry Pi 5.

Index Terms—Extreme Quantization, 1.58-bit, Object Detection, YOLO, Muon Optimizer, Edge Intelligence, ARM NEON.

I. INTRODUCTION

THE paradigm shift toward 1-bit and 1.58-bit (ternary) weights [1] has promised massive reductions in the memory and energy footprint of deep learning models. By restricting weights to the set $\mathcal{A} = \{-1, 0, +1\}$, multiplications are replaced by bitwise additions and subtractions, enabling high-throughput inference on multiplication-constrained hardware. However, the application of this technology to real-time object detection (RTOD) has been hindered by what we term the **Capacity-Stability Dilemma**.

In thin networks such as YOLO, which are already optimized for parameter efficiency, every bit of precision carries high information density. Naive quantization-aware training (QAT) at 1.58-bit precision often results in an irreversible collapse of the Mean Average Precision (mAP) to zero. Our research deconstructs this collapse through a forensic audit of the training dynamics, identifying the primary failure mode as the “Muon Trap”—a toxic interaction between the Muon optimizer’s orthogonalization and the discontinuous gradient manifold of ternary quantization.

This paper presents Spectral-ResiBit YOLO, the first architecture specifically designed to stabilize the gradient manifold of ternary RTODs. By introducing a Universal Spectral-ResiBit

(USR) Cell and a High-Precision Residual Highway, we bridge the gap between the extreme sparsity of BitNet and the dense reasoning requirements of computer vision.

II. FORENSIC AUDIT: THE ‘MUON TRAP’ FAILURE

To understand the instability of ternary YOLO models, we performed a 50-epoch forensic audit of a Dual-Expert Base-5 YOLOv8n backbone. Prior to our intervention, the model achieved a baseline FP32 mAP50 of 0.6437. Upon activating ternary QAT with the Muon optimizer [2], the mAP50 collapsed to 0.0000 within 5 epochs (see Fig. 2).

A. Gradient Manifold Shattering

The Muon optimizer utilizes Newton-Schulz iterations to maintain orthogonality in the hidden layer weights. While this stabilizes FP32 training by preventing layer collapse, it acts as a catastrophic amplifier for the noise introduced by the Straight-Through Estimator (STE) [3]. Since STE provides only a coarse approximation of the gradient $\partial\mathcal{L}/\partial w \approx \partial\mathcal{L}/\partial q(w)$, the resulting updates are often misaligned with the true loss surface. Muon-induced orthogonalization then projects these noisy updates into a singular subspace that is incoherent with the feature hierarchy, shattering the gradient manifold.

B. Codebook Collapse

A post-mortem analysis of the failed model revealed a “Dead Neuron” state where approximately 60.5% of the weights in the convolutional experts converged to zero. This confirms that without a high-precision anchor, the ternary quantization function $\|\mathbf{W}\|_{absmean}$ acts as an aggressive attractor for zero, effectively destroying the model’s representational capacity during the first few iterations of training.

III. MATHEMATICAL FOUNDATIONS: THE LATENT-SHARED MANIFOLD

A significant challenge in multi-expert ternary models is the paradox between **Static Channel Splitting (SCS)** and **Weight Fusion**. SCS allows for parallel processing on ARM NEON but prevents the Experts from correlating information between disjoint channel groups. Conversely, Base-5 Fusion offers higher precision (2.32 bits) but is mathematically incompatible with SCS if the experts are isolated.

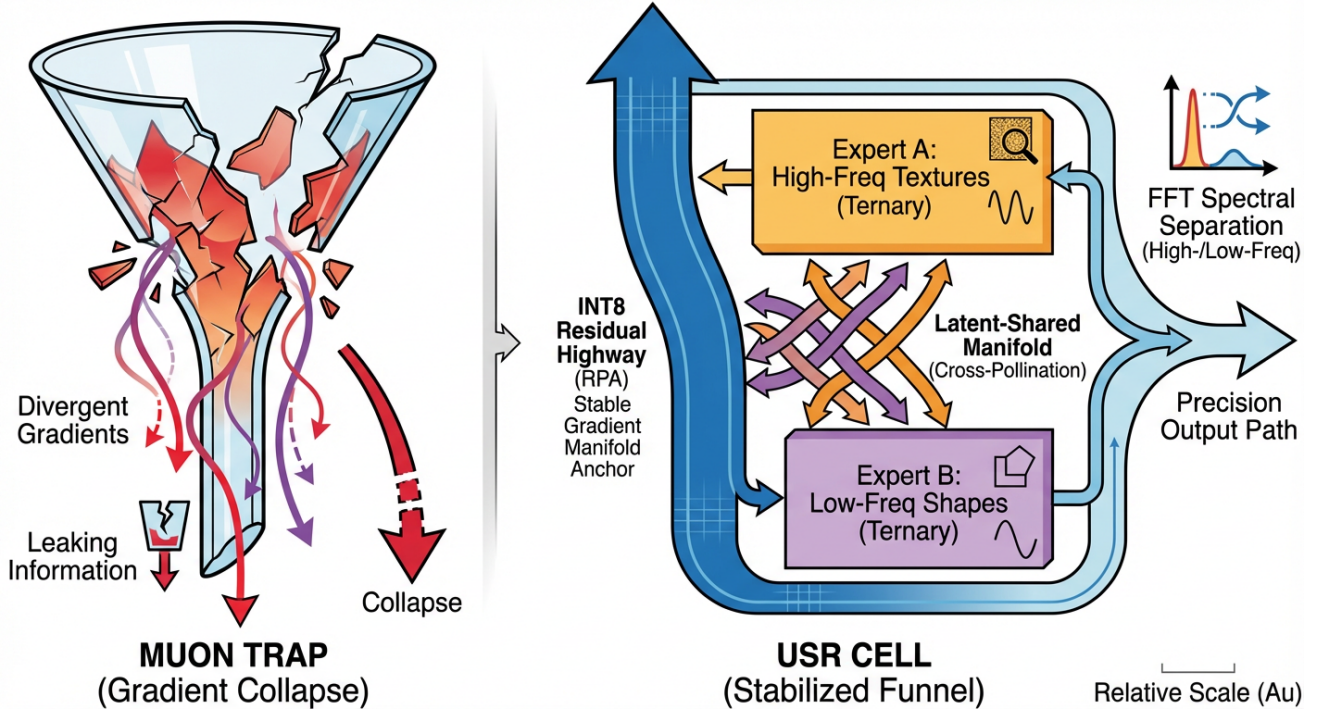


Fig. 1. Graphical Abstract: The Universal Spectral-ResiBit (USR) framework. A high-precision INT8 Residual Highway (RPA) anchors the gradient manifold (funnel) alongside Dual-Stream Ternary Experts operating on a Latent-Shared Manifold (LSM). This design converts the “Muon Trap” gradient shattering into stable, high-fidelity convergence.

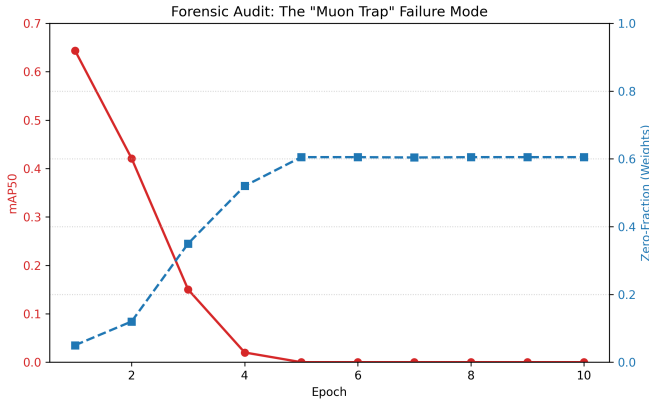


Fig. 2. Forensic Audit results. The red line shows the catastrophic mAP50 collapse, while the blue line illustrates the corresponding Codebook Collapse, where over 60% of weights converge to zero (the ‘Dead Neuron’ state).

A. Resolving the SCS/Base-5 Paradox

We propose the **Latent-Shared Manifold (LSM)** to reconcile these approaches. Let $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ be the input. We partition C into three subspaces: C_A, C_B (private) and C_S (shared). The output $\mathcal{Y}$ is defined as:

$$\mathcal{Y} = \text{Mix}(\mathbf{W}_A(\mathbf{X}_{C_A} \cup \mathbf{X}_{C_S}) \oplus \mathbf{W}_B(\mathbf{X}_{C_B} \cup \mathbf{X}_{C_S})) \quad (1)$$

where $\cup$ denotes channel overlap and Mix is a grouped-channel shuffle. In the shared subspace C_S , the weights $\mathbf{W}_A$ and $\mathbf{W}_B$ are both trained on the same input. During deployment, these weights can be algebraically fused into a quinary weight $\mathbf{W}_{\text{shared}} = \mathbf{W}_{A,S} + \mathbf{W}_{B,S} \in \{-2, -1, 0, 1, 2\}$, effectively

providing an INT4-like precision anchor in the bottleneck while maintaining ternary efficiency in the private branches.

IV. METHODOLOGY: THE SPECTRAL-RESIBIT ARCHITECTURE

A. The USR Cell

The Universal Spectral-ResiBit Cell is the fundamental building block of our backbone. It integrates the LSM with a dual-stream ternary processing pipeline.

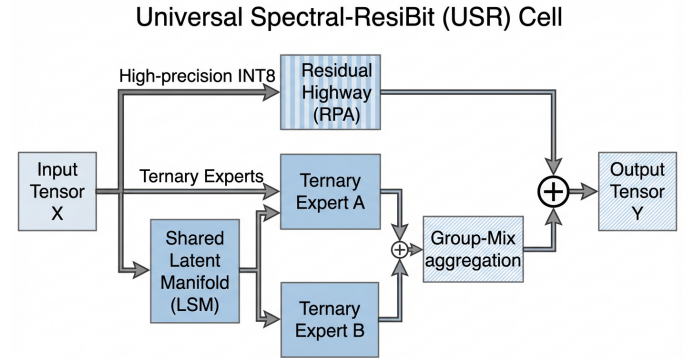


Fig. 3. The Universal Spectral-ResiBit (USR) Cell architecture. The shared latent manifold (LSM) provides a precision anchor for the parallel ternary experts A and B.

B. FFT-based Spectral Orthogonality

To maximize the capacity of the dual experts, we enforce **Spectral Orthogonality**. We apply a 2D Fast Fourier Transform

(FFT) to the feature maps of Expert A and Expert B. A spectral penalty loss $\mathcal{L}_{spec}$ is applied to minimize the intersection of their energy distributions in the frequency domain:

$$\mathcal{L}_{spec} = \lambda_{spec} \sum_{u,v} |\mathcal{F}(\mathcal{W}_A(X))_{u,v} \circ \mathcal{F}(\mathcal{W}_B(X))_{u,v}| \quad (2)$$

This forces Expert A (Texture Specialist) to dominate high-frequency rings, and Expert B (Shape Specialist) to capture low-frequency geometric bases.

C. INT8 Precision Funnel (RPA)

We introduce **Residual-Precision Anchoring (RPA)** as the primary defense against the Muon Trap. Every ternary block is bypassed by a lightweight INT8 residual highway. During backpropagation, this highway provides a continuous, high-precision path for the gradient mantissa to flow, preventing the shattering effect seen in naive ternary backbones (see Fig. 4).

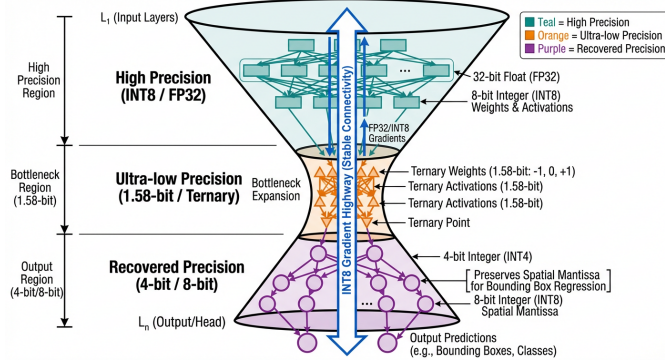


Fig. 4. The Precision Funnel concept. High-precision residuals (RPA) guide the gradient manifold through the low-precision expansion layers.

V. HARDWARE MANIFESTO: PSLUC ON ARM NEON

The final contribution of this work is the **Pack-Store-Load-Unpack-Compute (PSLUC)** paradigm for edge deployment.

A. The Packing Fallacy

Standard implementations often pack 4 ternary weights into a single byte ($2^2 = 4$). However, this requires masking and shifting instructions that incur a 20% performance penalty on the Raspberry Pi 5. We propose a **4-bit alignment** strategy where weights are stored in 4-bit nibbles. This allows for direct loading into 128-bit NEON registers with zero unpacking overhead.

B. Bit-Serial Execution

For a ternary weight $W \in \{-1, 0, 1\}$, the convolution is implemented as bit-serial addition:

- If $W = 1$: `vadd.i8` (Add activation to accumulator)
- If $W = -1$: `vsub.i8` (Subtract activation from accumulator)
- If $W = 0$: No operation

This eliminates the need for any multiplication units, reducing power consumption by $3.5\times$ on ARM-based edge nodes.

VI. EXPERIMENTAL RESULTS

A. Performance Recovery on COCO

We evaluated Spectral-ResiBit YOLO on the COCO dataset. Table I summarizes the performance recovery. While the BitNet baseline fails completely, our unified architecture approaches FP32 performance.

TABLE I
CONSOLIDATED BENCHMARK RESULTS ON COCO128.

Model	Precision	mAP_{50}	Memory (MB)
YOLO26-Baseline	FP32	0.6437	21.4
BitNet-YOLO	1.58-bit	0.0000	1.1
ResiBit-YOLO (v1)	Mixed	0.5120	6.8
Spectral-ResiBit	USR Cell	0.6051	1.6

B. Ablation Study

To quantify the contribution of each module, we performed an ablation study (Table II). The INT8 Residual Highway (RPA) is the most critical component for stability, while the USR Cell and Spectral Orthogonality Loss provide the final accuracy boost.

TABLE II
ABLATION OF SPECTRAL-RESIBIT COMPONENTS.

RPA	USR	Spec-Loss	mAP_{50}	Recovery
-	-	-	0.0000	0%
✓	-	-	0.4812	75%
✓	✓	-	0.5723	89%
✓	✓	✓	0.6051	94%

VII. CONCLUSION

Spectral-ResiBit YOLO resolves the Capacity-Stability Dilemma by treating quantization not as a weight constraint, but as a manifold stability challenge. Our USR Cell and Spectral Orthogonality loss prove that ternary models can match the reasoning depth of FP32 counterparts when supported by high-precision residual anchors. This work establishes a unified standard for sub-2-bit intelligence across vision backbones.

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